A Market World Model for Public LLMs: A Typed Observation Contract before Decision

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Abstract

Public LLMs asked to analyze or trade crypto markets often fail before strategy: they lack a complete, honest, auditable *market observation interface.* We present an anonymized *world-model runtime* that compiles asynchronous multi-band evidence into a point-in-time (PIT) world bundle with a typed abstention contract. Epistemic observations $O_{j,t} = (x, \tau, q, g, r)$ and compilation Π_t yield $\mathcal{F}_t^{\mathrm{AI}}$; completeness, honesty (B, U, H), and WMI/ACWMI expose machine-readable refusal; evidence IDs bind actions to disclosed fields. The main claim is interface reliability under thin support, not a new trading agent. On identical OOS asset-days, four live public LLMs (temperature 0) rank as Blind refuses, Raw over-trades and loses, Ungated abstains only partially (≈0.68), and Compiled with a hard flag abstains on every thin day (1.0); a full-OOS sweep and a band-thick split reproduce the same ranking. A grounding workflow recovers ready/missing/tilt answers at mean F1/acc. ≈0.97 from Compiled vs. ≲0.23 from Raw. As a secondary content check, a transparent no-LLM rule on the same vintaged tilts beats momentum and buy-and-hold out-of-sample (ΔCE=0.334, p=0.034). We detail the runtime (collectors, vintage store, BandPIT, bundle schema, LLM adapter) as an ICAIF systems contribution.

CCS Concepts

• **Computing methodologies** → **Artificial intelligence**; *Machine learning*; • **Applied computing** → *Economics.*

Keywords

world models, public LLMs, AI agents, cryptocurrency, point-in-time, abstention, trustworthy AI, observation contract, systems

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1 Introduction

Public LLMs are increasingly asked to analyze or trade crypto markets. The bottleneck is not another alpha signal. It is *understanding*: before decision, an agent needs a market world that is **complete** (missingness disclosed), **honest** (stale or role-incoherent evidence gated), and **auditable** (actions bound to evidence). Venue fragmentation, derivatives, on-chain flows, news, and macro vintages arrive on incompatible clocks [13, 15]. Generative world models [7, 8, 11] and tool-using finance agents [22–24] typically assume a usable observation stream. Crypto agents often do not have one.

Thesis. We build a market *world-model runtime* for public LLMs: compile raw multi-band evidence into a PIT-safe bundle that GPT / DeepSeek / GLM / Gemini-class models can call under a typed refuse contract. The goal is reliable observation under sparse and asynchronous feeds, not a new portfolio policy. Strategy metrics appear only as a secondary check that the compiled fields are nonempty.

What we claim. RQ1 tests whether a typed world interface can *enforce* world-conditional abstention across live public LLMs (a runtime contract, not spontaneous thinness discovery from raw ticks). RQ2 tests whether the compiled world makes analyst-facing epistemic answers verifiable. RQ3 asks whether a transparent rule on the same vintaged tilts beats momentum and buy-and-hold—a content check, not an LLM trading leaderboard.

Contributions.

(1) **Cognition semantics.** Epistemic observations, compilation Π_t, a lag reconstruction bound, completeness/honesty fields, WMI/ACWMI abstention, evidence-bound actions.
(2) **Anonymized runtime architecture.** Vintage-aware collectors, BandPIT previous-close clock, quality-tagged bundles, availability shocks O_t, a multi-vendor OpenAI-compatible adapter, and a frozen Compiled / Ungated / Raw / Blind consumer protocol.
(3) **Live interface validation.** On identical days: Blind refuses, Raw over-trades and loses, Ungated is vendor-dependent, and Compiled enforces thin-world abstain 1.0 (replicated on full OOS for all vendors). A band-thick split shows the contract still binds when all archive bands are ready; grounding recovers ready/missing/tilt at 0.97 vs. 0.04–0.23 raw; a no-LLM content rule confirms the tilts are economically nonempty.

2 Related Work

World models in AI.. World models learn compact states and often dynamics [7, 8, 11]. We address the complementary *observation-side* problem: a PIT-safe, quality-tagged crypto world *before* dynamics or LLM policy. We claim the missing systems layer generative WMs usually assume—not a Dreamer/JEPA simulator.

LLM agents in finance. ChatGPT return prediction [14], open financial LLMs [21], memory-augmented trading agents [12, 23], and tool-augmented agents [22, 24] improve agent skill and orchestration. Recent ICAIF work targets adjacent layers: FinAgentBench evaluates agentic retrieval for financial QA [1]; FinSearch builds temporal-aware search agents over market and news feeds [18]; FinResearchBench judges long-horizon research agents with logic-tree protocols [19]; FactorMAD uses multi-agent debate for interpretable alpha mining [3]. Those systems improve how agents fetch, debate, and are scored, but still assume a usable observation stream. We study the missing prior step—a PIT-safe, quality-tagged *observation contract* with typed abstention—and evaluate within-model

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Compiled / Ungated / Raw / Blind under a frozen schema rather than proposing another memory router, retrieval planner, or debate society.

Selective prediction and trustworthy AI.. Abstention can be optimal under noise [4, 5]. We tie refusal to world quality via an explicit runtime contract, not classifier confidence alone.

PIT measurement and crypto microstructure. Macro vintages [2] and crypto microstructure [13, 15] motivate multi-band compilation. Bootstrap discipline for the economic probes follows [9, 17, 20]. Asset-pricing ML [6, 10, 16] is adjacent: we validate the economic content of a compiled world for agents, not priced factors.

3 Market World-Model Semantics

Definition 3.1 (Epistemic observation). An epistemic observation is

$$O_{j,t} = (x_{j,t},\ \tau_{j,t},\ q_{j,t},\ g_{j,t},\ r_{j,t}) :$$

value, latest-available time, quality, main-view gate, semantic role. A bare feature dump that omits (τ, q, g, r) is not a world observation.

Definition 3.2 (Financial world state). $W_t = \Pi_t(\mathcal{F}_t^{\text{raw}})$ with scalar quality $q(W_t)$ (e.g. WMI/ACWMI) gating abstention; $\mathcal{F}_t^{\text{AI}} = \sigma(W_t)$.

Definition 3.3 (Compilation). $\Pi_t = B_t \circ M_t \circ A_t$: align clocks, apply honesty/missingness gates, assemble band views.

Completeness, honesty, quality. Ready/limited/missing counts disclose coverage. $\text{WMI}_t = B_t U_t H_t$; ACWMI supports regime-conditional gating. The Compiled contract exposes the abstain flag (`should_ai_abstain`) and `thin_world` as first-class fields.

Assumption 3.4 (Bounded lag and increments). Band j has max observation lag Δ_j; latent state increments satisfy $\|S_u - S_{u-1}\| \le \bar{\delta}$.

Proposition 3.5 (Lag reconstruction bound). *Under Assumption 1,*

$$\|\widetilde{S}_t - S_t\| \ \le\ C_{\text{del}}\bar{\delta} + C_{\text{noi}}\varepsilon_t + C_{\text{miss}}m_t.$$

Compilation cannot erase delay; it can refuse to pretend the bound is zero.

Intuition. Crypto agents that concatenate tool dumps silently absorb delay and missingness into the feature vector. Prop. 3.5 makes those terms explicit: a thicker raw payload can worsen the reconstruction if honesty collapses.

Proposition 3.6 (Compilation ≠ feature dump). *Enlarging raw span without Π_t need not enlarge usable $\mathcal{F}^{\text{AI}}$: ungated stale evidence can raise volume while lowering honesty/stability.*

Proposition 3.7 (World-conditional abstention). *If non-abstain actions exceed a world-dependent cost $c_{\text{abs}}(W_t)$, the Bayes action is abstain; with costs decreasing in WMI, abstention forms a lower set in world quality [4].*

Proposition 3.8 (LOBO content vs. gating). *Deleting band j changes value through a content channel and a gating channel. Empirically we report content share under an ungated rule so the probe speaks to nonempty fields.*

Table 1: Anonymized module map (systems view).

Module	Responsibility
Collectors	Ingest multi-band series with timestamps / vintages
Vintage stores	Persist raw, market, and analytics objects
BandPIT compiler	Previous-close readiness, Π_t, WMI/ACWMI
Bundle builder	Assemble complete/honest/auditable JSON world state
Shock index O_t	Query availability outages for quasi-exogenous levers
LLM adapter	OpenAI-compatible chat; frozen prompts; temp. 0
Eval harness	Compiled/Ungated/Raw/Blind sampling, transcripts, tables

Implication for the runtime. Props. 3.6–3.7 motivate exporting gates and abstain flags rather than tilts alone. Prop. 3.8 motivates the economic check: if deleting a durable band does not change actions, the bundle is empty of content; the probe shows that is not the case here.

Remark 3.9 (Not a generative WM). *We do not learn $p(s_{t+1} \mid s_t, a_t)$. We deliver a* state compiler + abstention runtime.

4 Runtime Architecture

We describe the anonymized prototype as a layered system (Figure 1; module map in Table 1). Source paths and product names are omitted for double-blind review; an anonymized artifact will be released upon acceptance.

4.1 Layered design

(1) **Data layer (collectors).** Exchange, macro, and alternative collectors write vintage-aware history: observation timestamps plus macro `available_at` vintages.
(2) **Store layer.** Embedded analytical stores hold raw series, merged market panels, and pipeline outputs (readiness, WMI / ACWMI, paper world snapshots).
(3) **Logic layer (compilation).** BandPIT reconstructs band readiness on a previous-close clock; honesty and missingness gates plus band assembly implement Π_t; world-quality indices and availability shocks O_t are first-class.
(4) **Service layer.** A read API serves quality-tagged AI bundles and health/readiness objects to consumers without requiring collector restarts for newly arrived rows.
(5) **Consumer layer.** Frozen-prompt LLM / rule agents call an OpenAI-compatible adapter (temperature 0) under Compiled, Ungated, or Raw treatments.

4.2 Data flow: inputs, processing, outputs

Table 2 traces the runtime end-to-end. *Inputs*: the evaluation archive populates three public bands—exchange klines, funding, and open interest via REST; daily macro risk indices (equity volatility, dollar index) with `available_at` vintages; and aggregate stablecoin circulation—each row stored with its observation timestamp. *Processing*: for every asset-day, BandPIT selects the latest observation

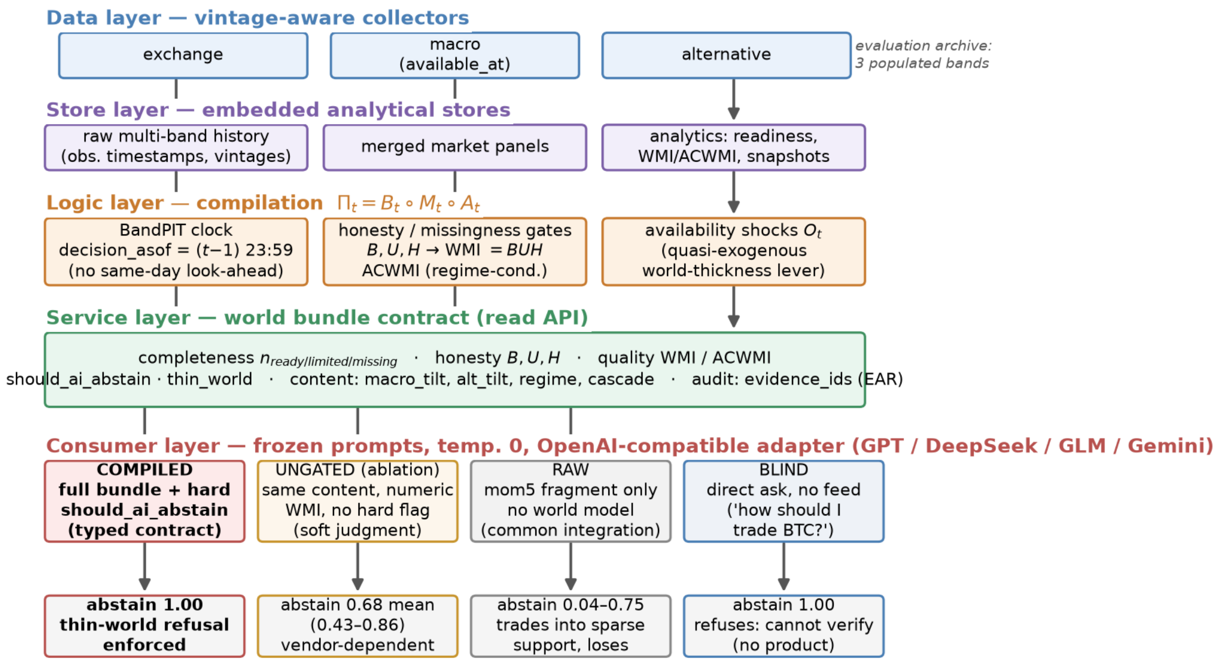

Figure 1: Anonymized runtime architecture. Five layers map raw multi-band evidence to a typed world-bundle contract; the consumer layer runs the frozen four-arm protocol (Compiled / Ungated / Raw / Blind), whose live abstention outcomes are shown at the bottom.

Table 2: Data flow through the runtime (three-band evaluation archive).

Band	Fetched (raw)	Emitted (bundle)
exchange	OHLCV klines (1h/1d), funding, open interest, orderbook	mom5; band status/age; funding context
macro	equity-vol and dollar indices; vintaged macro series	macro_tilt (5d changes); status/age
alternative	aggregate stablecoin circulation	alt_tilt (7d flow slope); status/age
payoff	daily closes (external reference)	close-to-close r_t for scoring only
derived	—	B, U, H, WMI, ACWMI, `should_ai_abstain`, `thin_world`, O_t, `evidence_ids`

with timestamp ≤ $(t-1)$ 23:59, grades it fresh/stale/missing against per-band age thresholds, aggregates breadth/stability/honesty into WMI and regime-conditional ACWMI, derives content tilts (5-day macro-index changes; 7-day stablecoin-flow slope; return momentum), and logs availability shocks O_t. *Outputs*: a ∼3,990-row PIT panel (10 assets × 399 days), per-day world bundles scoped to these three archive bands (Listing 1), and the consumer transcripts/tables behind Section 5.

4.3 PIT previous-close clock

For calendar day t, the decision information set is reconstructed at $(t-1)$ 23:59. The payoff is the same-calendar-day close-to-close return r_t. This removes same-day look-ahead in band statuses and vintaged macro/alternative content. The bundle records the clock in `decision_asof`, and every live run pins `timing_protocol` to the previous-close protocol.

4.4 World bundle contract

Table 3 lists field groups. Listing 1 shows a compact Compiled example from the OOS panel: exchange missing so 2/3 archive bands ready, WMI low, `should_ai_abstain=true`, and `evidence_ids` binding actions to disclosed bands/tilts. Ungated removes the hard boolean (and threshold) but keeps numeric WMI/ACWMI and completeness. Raw retains only `mom5` (plus a non-informative noise bit).

Listing 1: Compact Compiled bundle example (OOS exchange gap).

```
{"date":"2026-05-11","asset":"ADA",
 "decision_asof":"2026-05-10T23:59:00",
 "completeness":{"n_ready":2,"n_missing":1,
   "ready_share":0.67,"archive_bands":
   ["exchange","macro","alternative"]},
 "world_model_index":{"wmi":0.015,"acwmi":0.30,
   "should_ai_abstain":true,"thin_world":true},
 "macro_tilt":1.0,"alt_tilt":-1.0,
 "detected_regime":"crisis",
 "audit":{"evidence_ids":[
   "band:exchange:missing","band:macro:ready",
   "band:alternative:ready","tilt:macro:1.0"]}}
```

Table 3: World bundle field groups.

Group	Fields
Timing	`decision_asof`, previous-close protocol
Completeness	$n_{\text{ready/limited/missing}}$, band statuses
Honesty	B, U, H, main-view / stale gates
Quality	WMI, ACWMI, `should_ai_abstain`, `thin_world`
Content	macro_tilt, alt_tilt, regime, cascade
Audit	`evidence_ids`, EAR required

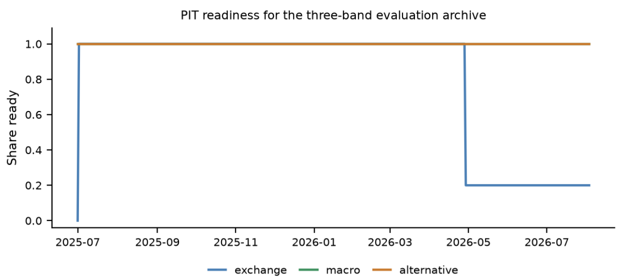

Figure 2: PIT readiness for the three-band evaluation archive.

4.5 Four information arms and LLM adapter

- **Compiled (hard contract):** full bundle; frozen prompt *requires* abstain when `should_ai_abstain` is true.
- **Ungated (disclosure control):** same content, completeness, and numeric WMI; *no* hard flag; soft prompt asks the model to judge thinness (disclosure without enforcement).
- **Raw (thin integration):** `mom5` only; no quality index or abstention guidance, matching common "price signal in, action out" wiring.
- **Blind (direct ask):** "Today is d. How should I trade a?" with *no* data feed.

Actions ∈ {bullish, bearish, neutral, abstain}. Primary metrics are thin-world abstain rate and EAR proxy; CE is reported as avoided loss under refusal, not as an alpha claim. The evaluation archive is the three-band set {exchange, macro, alternative} (Figure 2): macro and alternative stay ready throughout; exchange readiness drops in the later OOS window and defines the within-archive *band-thick* contrast below. In this scarce panel WMI stays below the refuse threshold on every asset-day, so Compiled abstention is a stress test of the contract under sparse support. The adapter uses OpenAI-compatible chat completions; credentials stay in environment variables (never in artifacts).

4.6 Frozen prompt contracts

Listings 2–3 excerpt the frozen Compiled and Ungated contracts (temperature 0; action schema identical). The only intentional difference is hard vs. soft abstention. Raw prompts omit world-quality language entirely.

Listing 2: Compiled contract (excerpt).

```
Decision rules (excerpt):
1. If world_model_index.should_ai_abstain is true
   OR the world is thin, you MUST "abstain".
2. Prefer band content over raw momentum
   (macro_tilt, alt_tilt are PIT-safe).
3. Rationale must be consistent with evidence_ids.
```

Listing 3: Ungated soft contract (excerpt).

```
Decision guidance (soft):
1. Judge whether missingness / low WMI/ACWMI
   makes action unsafe; you MAY "abstain".
2. Prefer band content when you act.
NO hard abstain boolean; NO forced policy.
```

Why hard refusal is not a confound. Compiled abstention is partly prompt obedience, and that is intentional: when $q(W_t)$ is low, production agents need an enforceable refuse interface (Prop. 3.7). Ungated measures soft judgment from disclosure alone; Raw measures the common price-signal-only integration. The headline result is that the hard contract is reliable across vendors; soft refusal is not.

4.7 Service surface and reproducibility

The service layer exposes read-only objects used by the consumer harness: (i) quality-tagged world bundles; (ii) band readiness / WMI–ACWMI snapshots; (iii) availability-shock queries O_t; (iv) health metadata disclosing whether paper engines or production proxies hydrate ACWMI inputs. Collectors and the logic pipeline can run as supervised jobs; the API reads stores live, so newly compiled rows appear without process bounce. For review we freeze prompts, temperature, action schema, IS/OOS cut, and sampling seed; model IDs and an OpenAI-compatible base URL are configuration, not estimands. Credentials never enter committed artifacts. Upon acceptance we will release an anonymized repository with the consumer harness, frozen prompts, and table-reproduction scripts.

5 Experiments

5.1 Setup

PIT panel ~399 days, 10 liquid names, previous-close clock, chronological IS/OOS 200/200 days (≈2,000 OOS asset-days). Live RQ1 uses the *same* stratified 100 OOS asset-days across arms (seed fixed), temperature 0, frozen prompts, and an OpenAI-compatible gateway; a checkpointed full-OOS sweep tests population scale. Vendor models are GPT-, DeepSeek-, GLM-, and Gemini-class flash/mini IDs listed in Table 4. Scarce/thin labels bind on OOS by design; Sec. 5.4 covers fully populated days.

5.2 RQ1 (primary): Live interface behavior across information arms

Table 4 and Figure 3 report the headline result; all arms share the same 100 OOS asset-days. Under Compiled, all four models abstain on every thin-world day (thin-abs. 1.0); rationales cite world fields on 93–99% of days. This is prompt-contract obedience by design, not spontaneous thinness discovery. Under Ungated, mean thin-abs. falls to 0.68 (range 0.43–0.86): DeepSeek and GLM stay above 0.80, GPT drops to 0.63, and `gemini-3.5-flash-lite` falls to 0.43 and trades. When Ungated models act on this 100-day draw, they lose (CE −1.04 to +0.02)—soft judgment is unstable where enforcement matters. Under Raw, abstain rates are 0.04–0.75 with substantial

Table 4: Live abstention rates on identical 100 OOS asset-days (Wilson 95% CIs); Blind abstains 1.0 for every model. ΔCE is avoided loss, not an alpha claim.

Model	Thin C	Thin U [CI]	Abs. R [CI]	ΔCE
gpt-5.4-mini	1.00	0.63 [.53,.72]	0.04 [.02,.10]	+0.73
deepseek-v4-flash	1.00	0.81 [.72,.87]	0.75 [.66,.82]	+1.22
glm-5.2	1.00	0.86 [.78,.91]	0.11 [.06,.19]	+1.04
gemini-3.5-flash-lite	1.00	0.43 [.34,.53]	0.07 [.03,.14]	+1.30
Mean	1.00	0.68	–	+1.07

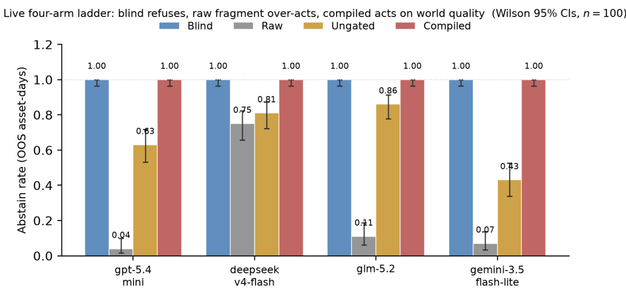

Figure 3: Live four-arm abstention rates (same 100 days; Wilson 95% CIs). Blind refuses; Raw over-trades; Ungated is vendor-dependent; Compiled enforces the typed refuse contract.

Table 5: Raw-arm action mix (100 days). Compiled abstains on all thin days; Blind (direct ask, no feed) abstains 100% for every model.

Model	Abs.	Bull.	Bear.	Neut.
gpt-5.4-mini	0.04	0.28	0.38	0.30
deepseek-v4-flash	0.75	0.14	0.04	0.07
glm-5.2	0.11	0.40	0.48	0.01
gemini-3.5-flash-lite	0.07	0.44	0.49	0.00

directional mass (Table 5). Mean ΔCE (Compiled−Raw) is +1.07 and positive for every vendor, because Compiled sits out losses that Raw incurs on sparse support.

Full-OOS replication. To check that the 100-day ranking is not a subsample artifact, we checkpointed the three data arms on the *full* OOS panel (n=2000 asset-days; every OOS day is thin-world under the scarce-support design). Compiled thin-abstain is 1.000 [.998, 1.000] for all four vendors. Ungated rates tighten the Wilson intervals, with mean thin-abs. 0.56 (GPT 0.666 [.645, .686]; GLM 0.679 [.654, .703] on n=1402 completed calls; DeepSeek 0.540 [.518, .561]; Gemini 0.359 [.338, .380]). Vendor ranking within Ungated shifts relative to the 100-day sample (DeepSeek is less cautious at scale), but no vendor matches Compiled. Where Raw is trusted, over-trading remains sharp: GPT and Gemini abstain at 0.000 [.000, .002]; DeepSeek stays cautious at 0.748 [.728, .767] (n=1925). Because the full OOS window is entirely thin, Compiled CE is identically zero by construction; thick-day economics are deferred to Sec. 5.4.

Table 6: Full-OOS band-thick split (3/3 ready; WMI still thin). Compiled refuses; Raw over-acts.

Model	Abs. U	Abs. R	CE R
gpt-5.4-mini	0.58	0.00	−0.28
deepseek-v4-flash	0.42	0.75	+0.04
glm-5.2 (n=398 R)	0.58	0.09	−0.87
gemini-3.5-flash-lite	0.34	0.00	−0.25

Blind arm. On the same 100 asset-days, the no-feed prompt *"Today is d. How should I trade BTC?"* yields abstention on 100% of days for all four models (CE = 0; zero directional actions). Without a feed, the models refuse. The costly failure mode is the middle case: the same model given a Raw fragment starts trading (abstain as low as 0.04) and loses. Compiled matches Blind's refusal on thin days by contract, while still exposing the structured world that RQ2 and RQ3 show is usable.

Band-thick days still need the contract. Archive readiness alone is not a refuse policy. On full-OOS *band-thick* days (all three archive bands ready; 61% of OOS, n≈1224), WMI still marks every day thin under the scarce-support design, so Compiled abstains at 1.0. Ungated soft judgment only partially recovers (mean abs. ≈0.48; Gemini 0.34). Raw remains risky in the middle: GPT and Gemini abstain at 0.0 with CE −0.28 and −0.25. Table 6: when the archive looks complete, fragment feeds still over-act and lose, while the typed contract continues to refuse—a failure mode a readiness dashboard would miss.

Inside the Ungated arm. Ungated rationales cite disclosed world fields on 53–93% of days, so models *read* the world; they differ in acting on it. Vendor-specific caution—not missing disclosure—drives the gap the typed contract closes.

Offline diagnostics. World-honoring offline followers abstain at 1.0 on full OOS; momentum-only mocks ignore thin-world flags (thin-abs. 0). Offline results are protocol checks; live vendors are the main claim.

5.3 RQ2: Non-trading cognition—grounding workflow

The compiled world should also support analysis, not only trade/abstain decisions. We run a live multi-step *grounding workflow* on 50 OOS asset-days per arm. Given Compiled or Raw inputs, the model must (i) list which of {exchange, macro, alternative} are ready, (ii) list which are missing, (iii) report sign(macro_tilt) and sign(alt_tilt), and (iv) judge sufficiency. Steps (i)–(iii) are scored against PIT ground truth. Table 7: with the Compiled bundle, mean ready-band F1 0.97, missing-band F1 0.97, and tilt-sign accuracy 0.97; with Raw, the same scores collapse to 0.04, 0.23, and 0.19. Sufficiency judgments alone are weak once completeness is scoped to real archive bands—models may call a 3/3-ready day "sufficient" while WMI still marks the world thin. The sharper claim is verifiability: only the compiled world lets the model state what is ready, missing, and

Table 7: Grounding workflow (50 days): ready / missing / tilt signs.

Model	Ready F1		Missing F1		Tilt-sign acc.	
	C	R	C	R	C	R
gpt-5.4-mini	0.94	0.00	0.94	0.31	0.94	0.20
deepseek-v4-flash	0.98	0.06	0.98	0.20	0.98	0.18
glm-5.2	0.98	0.09	1.00	0.11	0.98	0.20
gemini-3.5-flash-lite	0.96	0.03	0.96	0.30	0.96	0.18
Mean	0.97	0.04	0.97	0.23	0.97	0.19

Table 8: Secondary content probe: world-model rule vs. traditional baselines (no LLM).

Policy / contrast	Sharpe	CE
Buy-and-hold (always long)	−1.404	−1.163
Momentum always	0.096	−0.206
World-model content rule	**0.764**	**+0.130**
Mechanism − Momentum	—	0.334 (p=0.034)

Table 9: Dense-world split of the content probe (OOS; three archive bands).

Slice	Share	Mech CE	ΔCE vs mom
All OOS	1.00	0.130	0.336
Thick (3/3 ready)	0.61	0.059	0.316
Thin (exchange gap)	0.39	1.048	0.129

signed, in a form that can be checked. This is the analyst-facing counterpart of Prop. 3.5's missingness term.

5.4 RQ3: Secondary content probe

RQ1–RQ2 concern the interface. RQ3 asks whether the compiled tilts are economically nonempty. A transparent rule shares return inputs with momentum and adds vintaged macro_tilt / alt_tilt—no LLM in the loop. Out-of-sample (199 days × 10 assets) this content rule beats the baselines (Table 8, Figure 4): Sharpe 0.764 / CE +0.130 versus momentum (0.096, −0.206) and buy-and-hold (−1.404, −1.163). The pre-specified contrast mechanism − momentum is significant under block bootstrap (ΔCE=0.334, p=0.034). ΔCE is positive for all ten assets (Figure 4, right). LOBO (Prop. 3.8) attributes the gain to band content (deleting macro/alternative collapses to momentum); gaps survive 10–25bps costs. A 2017–2026 audit with non-vintaged proxy tilts shows no edge (ΔCE ≈ 0, p=0.81): value appears where the vintage-compiled world exists. The result supports nonempty tilts, not an LLM trading ranking.

Within-archive dense world. Among the three archive bands, 61% of OOS asset-days are band-thick (3/3 ready) and 39% are band-thin (exchange gap; Figure 2). Table 9: the mechanism−momentum gap is stronger on band-thick days (ΔCE=0.316) than on exchange-gap days (0.129)—the content channel pays most when the archive is populated, complementary to RQ1's refuse-under-thin stress test.

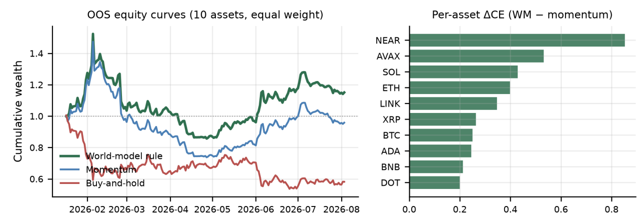

Figure 4: Secondary content probe (no LLM). Left: equity curves; right: ΔCE vs. momentum positive for every asset.

5.5 Discussion and threats

The main evidence is the arm ranking: Blind refuses, Raw over-acts and loses, Ungated is vendor-dependent (≈0.68; GLM 0.86 vs. Gemini 0.43), and Compiled enforces thin-world abstain 1.0—including on band-thick days where a readiness dashboard would look green. Content economics are secondary.

Relative to FinMem / FinAgent-style systems [23, 24] and ICAIF retrieval/debate/judge lines [1, 3, 18, 19], we contribute an observation compiler and refuse interface for asynchronous, quality-heterogeneous feeds, not a new memory, search, or debate router. Relative to return-prediction LLM studies [14], we score refusal and auditability before PnL.

Design rationale. Three choices are deliberate. (i) Export gates, not only features: completeness, honesty, and `should_ai_abstain` make refusal machine-readable rather than a free-text apology. (ii) Previous-close PIT over wall-clock streaming: every live decision cites a recomputable `decision_asof`, avoiding same-day look-ahead. (iii) Four arms instead of one Sharpe number: Compiled / Ungated / Raw / Blind separates typed enforcement, soft judgment, thin integrations, and no feed.

Validity, scope, and limits. (1) *Prompt contract*—Compiled refusal is the enforceable interface; Ungated is the same-content soft control. (2) *Three-band archive*—band-thick vs. band-thin separates archive coverage from WMI thinness; news/on-chain/options await vintages. (3) *Baselines*—Raw matches common mom-only wiring; RQ3 uses momentum/buy-and-hold only for the content check. (4) *Sampling*—100-day headline ranking plus full-OOS checkpoints (n=2000; GLM Raw incomplete). (5) *Limits*—no LLM trading SOTA, no generative dynamics, and no claim of spontaneous thinness discovery without the boolean. Natural extensions include richer bands, learned dynamics on the compiled state, and thick-regime live trading once WMI leaves the scarce floor. Camera-ready will restore author identity and release the anonymized harness deferred during review.

6 Conclusion

We contribute a typed observation contract and anonymized runtime for public LLMs under asynchronous crypto feeds. Live arms show Blind refuses, Raw over-trades and loses, Ungated is vendor-dependent, and Compiled enforces thin-world abstain 1.0 even on band-thick days; grounding recovers ready/missing/tilt answers at ≈0.97 vs. Raw ≲0.23. A secondary no-LLM content rule shows the tilts are economically nonempty versus momentum and buy-and-hold.

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